

Software Project Management

Estimation Experience and Beware of the Sales Guy...



Introduction

- Many software projects fail:
 - due to faulty project mgmt. practices:
 - It is important to learn different aspects of software project management.
- Goal of software project management:
 - enable a group of people to work efficiently towards successful completion of a software project.
- It is important to know the key role of project managers.

Responsibility of project managers

- Project proposal writing,
- Project cost estimation,
- Scheduling,
- Project staffing,
- Project monitoring and control,
- Software configuration management,
- Risk management,
- Managerial report writing and presentations.

Project Manager

- Project manager's activities are varied.
 - can be broadly classified into two:
 - project planning,
 - Undertaken - Once a project is found to be feasible,
 - project monitoring and control activities.

Sliding Window Planning

- Involves project planning over several stages:
 - protects managers from making big commitments too early.
 - More information becomes available as project progresses.
 - Facilitates accurate planning

SPMP Document

- After planning is complete,
 - Document the plans,
 - Result: A Software Project Management Plan (SPMP) document.

Organization of SPMP Document

- **Introduction** (Objectives, Major Functions, Performance Issues, Management and Technical Constraints)
- **Project Estimates** (Historical Data, Estimation Techniques, Effort, Cost, and Project Duration Estimates)
- **Project Resources Plan** (People, Hardware and Software, Special Resources)
- **Schedules** (Work Breakdown Structure, Task Network, Gantt Chart Representation, PERT Chart Representation)
- **Risk Management Plan** (Risk Analysis, Risk Identification, Risk Estimation, Abatement Procedures)
- **Project Tracking and Control Plan**
- **Miscellaneous Plans** (Process Tailoring, Quality Assurance)

Estimation

- “The single most important task of a project: setting realistic expectations.

Unrealistic expectations based on inaccurate estimates are the single largest cause of software failure.”

Futrell, Shafer and Shafer, “Quality Software Project Management”

Why its important to you!

- Program development of large software systems normally experience 200-300% cost overruns and a 100% schedule slip.
- 15% of large projects deliver...NOTHING!
- Key reasons...poor management and inaccurate estimations of development cost and schedule.
- If not meeting schedules, developers often pay the price!

Project planning

- Requires utmost care and attention - commitments to unrealistic time and resource estimates result in:
 - irritating delays.
 - customer dissatisfaction
 - adverse affect on team morale
 - poor quality work
 - project failure.

Remember: **A Project Manager should never OVER-COMMIT**

What is the solution?

What are The Problems?

- Predicting software cost
- Predicting software schedule
- Controlling software risk
- Managing /tracking project as it progresses

Fundamental questions

- How much effort is required to complete an activity?
- How much calendar time is needed to complete an activity?
- What is the total cost of an activity?
- Project estimation and scheduling are interleaved management activities.

Software Cost Components

- Hardware and software costs.
- Travel and training costs.
- Effort costs (dominant factor in most projects)
 - The salaries of engineers involved in the project;
 - Social and insurance costs.
- Effort costs must take overheads into account
 - Costs of building, cooling, lighting.
 - Costs of networking and communications.
 - Costs of shared facilities (e.g library, staff restaurant, etc.).

Costing and Pricing

- Estimates are made to discover the cost, to the developer, of producing a software system.
- There is not a simple relationship between the development cost and the price charged to the customer.
- Broader organisational, economic, political and business considerations influence the price charged.

Software Pricing Factors

Market opportunity	A development organisation may quote a low price because it wishes to move into a new segment of the software market. Accepting a low profit on one project may give the opportunity of more profit later. The experience gained may allow new products to be developed.
Cost estimate uncertainty	If an organisation is unsure of its cost estimate, it may increase its price by some contingency over and above its normal profit.
Contractual terms	A customer may be willing to allow the developer to retain ownership of the source code and reuse it in other projects. The price charged may then be less than if the software source code is handed over to the customer.
Requirements volatility	If the requirements are likely to change, an organisation may lower its price to win a contract. After the contract is awarded, high prices can be charged for changes to the requirements.
Financial health	Developers in financial difficulty may lower their price to gain a contract. It is better to make a smaller than normal profit or break even than to go out of business.